A W-LOCAL MORPHISM BETWEEN W-LOCAL SPECTRA WHICH IDENTIFIES LOCAL RINGS AND IS BIJECTIVE ON COMPONENTS IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the bijection on the underlying spaces from the bijection on closed points and the decomposition of a w-local space into local rings at closed points, the comparison of the localizations through flatness and going down, and the closing local criterion for a ring map to be an isomorphism.

Editorial note. The label lemma-w-local-morphism-equal-points-stalks-is-iso names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-w-local-morphism-equal-points-stalks-is-iso –start 1154 –end 1182 –local-id 28000 returned RESULT=OK (L28000 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. This card opens the second numbering segment L28000-L29999 (the first segment L3400-L3599 was closed by L3599); per LANE_BRIEF section 4 already registered cards keep their numbers.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Going down for flat ring maps. Algebra, the lemma that a flat ring map satisfies going down. Supplies the going down property of the flat map A -> B used to compare the localizations.
- Uniqueness of the prime below for flat maps between localizations. Algebra, the lemma that for a flat map with going down the primes lying over a given prime are related as required, giving the identification of the localizations of B. Supplies the conclusion that for any prime q of B lying over p in A one has B_q = B_p.
- Characterization of zero by vanishing of the localizations. Algebra, the lemma that a ring map is an isomorphism if it induces isomorphisms after localizing at every prime. Supplies the final step concluding A = B from the equality of all localizations.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HN: sources/stacks-chow/tags.txt, source lines 649–649. Prerequisite pointer invoked at proof line(s) 1181; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00HS: sources/stacks-chow/tags.txt, source lines 654–654. Prerequisite pointer invoked at proof line(s) 1174; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00EA: sources/stacks-chow/tags.txt, source lines 527–527. Prerequisite pointer invoked at proof line(s) 1175; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A \to B$ be ring map such that*

(1) *$A \to B$ identifies local rings,*
(2) *the topological spaces $\mathrm{Spec}(B)$, $\mathrm{Spec}(A)$ are w-local,*
(3) *$\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is w-local, and*
(4) *$\pi_0(\mathrm{Spec}(B)) \to \pi_0(\mathrm{Spec}(A))$ is bijective.*

Then $A \to B$ is an isomorphism

Proof. Let $X_0 \subset X = \mathrm{Spec}(A)$ and $Y_0 \subset Y = \mathrm{Spec}(B)$ be the sets of closed points. By assumption Y_0 maps into X_0 and the induced map $Y_0 \to X_0$ is a bijection. As a space $\mathrm{Spec}(A)$ is the disjoint union of the spectra of the local rings of A at closed points. Similarly for B. Hence $X \to Y$ is a bijection. Since $A \to B$ is flat we have going down (Algebra, Lemma 00HS). Thus Algebra, Lemma 00EA shows for any prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p} \subset A$ we have $B_\mathfrak{q} = B_\mathfrak{p}$. Since $B_\mathfrak{q} = A_\mathfrak{p}$ by assumption, we see that $A_\mathfrak{p} = B_\mathfrak{p}$ for all primes $\mathfrak{p}$ of A. Thus $A = B$ by Algebra, Lemma 00HN. $\square$